

The transmission line

ELEC0447 - Analysis of electric power and energy systems

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Overview

1. Introduction
2. Distributed model
3. Surge impedance loading
4. Lumped transmission line model
5. Line rating

Introduction

The background of the slide features a minimalist design with teal-colored geometric shapes. Two large teal triangles point upwards from the bottom corners, meeting at a central point. Below this meeting point, a smaller, darker teal triangle points downwards. The remaining space is white, providing a clean backdrop for the text.

What will we learn today?

- ▶ The transmission line
- ▶ An introduction to power flow analysis

You will be able to do exercises 4.3, 4.4, 4.6, 4.7, 4.8, 4.9, 4.10, 4.11, 4.12, Lab4 (power-flow in python), 5.1, 5.2, 5.5, 5.6 from Ned Mohan's book.

Introduction video link

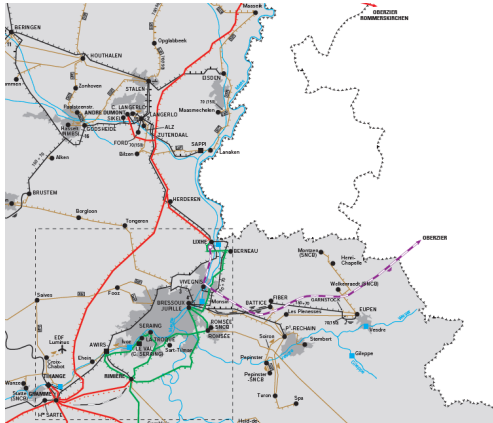
Definition

- ▶ An (overhead) transmission **line** is a set of 3 bundles of conductors corresponding to the three phases of the system.
- ▶ Commonly used **voltages** range from 70 kV to 380 kV in Belgium (more where distances are larger).
- ▶ Minimum **distances** between conductors depend on the voltage level, and thus electrical properties also depend on the voltage level.
- ▶ Underground **cables** are more and more used. They can be modeled in a similar way as overhead transmission lines, but they have different properties.



A pylon and lines from SwissGrid
(picture by B. Cornélusse).

A part of ELIA's network



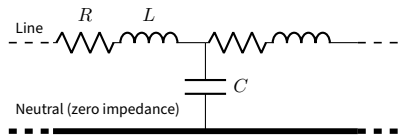
LIGNES AERIENNES		BOVENGRONDSE LIJNEN	
Tension d'exploitation 380kV 220kV 150kV 110kV 70kV		Uitvoeringspanning 380kV 220kV 150kV 110kV 70kV	
Nombre de terres prises 1 1 2 1 2 2 > 2		Aantal draadstellen voorsien uitgerust 1 1 2 1 2 2 > 2	
(voir légende de référence dans le tableau des composants)		(voir références pour les lignes de tension dans le tableau des composants)	
en construction ou en projet 2 ^{ème} terme en construction ou en projet		in aanbouw of in ontwerp 2 ^{de} draadstel in aanbouw of in ontwerp	
lignes à 2 terres de sections différentes tension d'exploitation inférieure à la tension de construction ligne appartenant à un tiers (T)		lijn met 2 draadstellen van verschillende spanningen uitvoeringspanning lager dan de constructiespanning lijn eigendom van een derde (T)	
Tableau des compositions des lignes à plus de 2 terres:		Samenstellingstabel van de lijnen met meer dan 2 draadstellen:	
1 2 x 150 + 1 x 70 (4 x 150)	7 2 x 300 (4 x 300)	13 1 x 70 (4 x 150)	
2 4 x 150	8 2 x 70	14 1 x 220 + 2 x 70	
3 1 x 150 + 2 x 70 (4 x 150)	9 2 x 110 (4 x 150)	15 2 x 150 (4 x 150)	
4 1 x 150 + 2 x 70 (4 x 150)	10 1 x 150 + 1 x 70 (4 x 150) + 1 x 70	16 2 x 150 (4 x 150)	
5 2 x 150 + 2 x 70 (4 x 150)	11 4 x 70	17 2 x 230	
6 2 x 150 (2 x 300 + 2 x 150)	12 2 x 150	18 1 x 300 + 2 x 150 (2 x 300 + 2 x 150)	

CABLES SOUTERRAINS		ONDERGRONDSE KABELS	
380kV 220kV 150kV 110kV 70kV HVDC		380kV 220kV 150kV 110kV 70kV HVDC	
en construction ou en projet câbles parallèles câble appartenant à un tiers (T)		in aanbouw of in ontwerp parallelle kabels kabel eigendom van een derde (T)	

Source: <https://www.elia.be/fr/infrastructure-et-projets/nos-infrastructures>

Transmission line parameters

A *chunk* (a tiny piece) of transmission line can be represented as:



with R , L and C expressed **per unit of length**.

where

- ▶ R represents the series resistance, as small as possible to minimize RI^2 (influence of the frequency and skin effect)
- ▶ the series inductance L models the magnetic coupling between phases
- ▶ the shunt capacitance C models the capacitive coupling between phases
- ▶ a shunt conductance G can be added to model e.g. the leakage current through insulators

Typical cross section of a high voltage overhead conductor

Skin depth (depth at which decay of current density is $1/e$ of surface density):

Material	Frequency [Hz]	depth [mm]
Copper	50	9.4
Copper	60	8.6
Aluminum	50	12.0
Aluminum	60	10.9



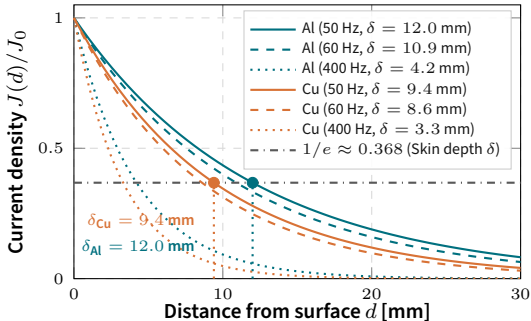
Aluminium Conductor
Steel-Reinforced (ACSR)

Skin effect: current density distribution

Current density decays with depth d from surface:

$$J(d) = J_0 e^{-d/\delta}, \quad \delta = \sqrt{\frac{\rho}{\pi f \mu}}$$

- ▶ At depth $d = \delta$: $J(\delta) = J_0/e \approx 36.8\%$.
- ▶ Within 3δ : $\approx 95\%$ of current is carried.
- ▶ Higher f or $\sigma \implies$ smaller δ .
- ▶ Effective AC resistance $R_{AC} > R_{DC}$.
- ▶ **ACSR**: steel core in center carries negligible current, providing mechanical strength without losses!



Approximate Overhead Transmission Line Parameters

For bundled conductors at 60 Hz.

Nominal Voltage	R (Ω/km)	ωL (Ω/km)	ωC ($\mu\text{F}/\text{km}$)
230 kV	0.06	0.50	3.4
345 kV	0.04	0.38	4.6
500 kV	0.03	0.33	5.3
765 kV	0.01	0.34	5.0



Approximate underground cable parameters

For conductors at 60 Hz.

Nominal Voltage	R (Ω/km)	ωL (Ω/km)	ωC ($\mu\text{F}/\text{km}$)
110 kV	0.05	0.030	95
220 kV	0.04	0.025	115
330 kV	0.03	0.020	130
400 kV	0.02	0.018	150

CU/XLPE/PVC/STA/PVC



CU/XLPE/PVC



CU/XLPE/PVC/SWA/PVC



AL/XLPE/PVC

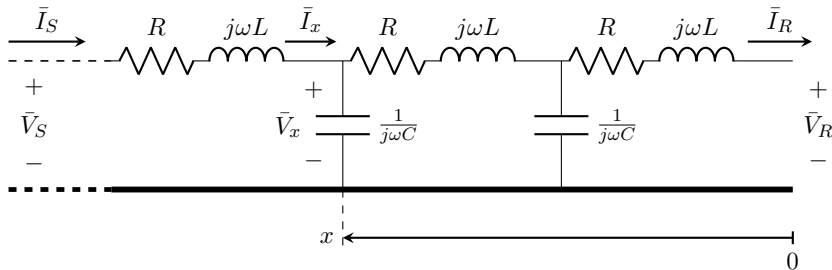


Distributed model

The background of the slide features a minimalist design with teal-colored geometric shapes. Two large teal triangles point upwards from the bottom, meeting at a central point. A smaller, darker teal triangle is positioned at the bottom center, overlapping the base of the two larger triangles. The text "Distributed model" is centered in the white space above the triangles.

Distributed parameter representation I

We consider that we are in sinusoidal steady state. On a per-phase basis, the line can be represented as many chunks connected to each other:



Distributed parameter representation II

How do voltage and current evolve as a function of the position on the line?

Let's fix the origin of x at the receiving end where $\bar{V}_R$ and $\bar{I}_R$ are assumed known.

- ▶ As R is small, R is considered as *lumped* (a discrete resistive element on one side of the line)
- ▶ $\frac{d\bar{V}(x)}{dx} = j\omega L\bar{I}(x)$
- ▶ $\frac{d\bar{I}(x)}{dx} = j\omega C\bar{V}(x)$

Hence

$$\frac{d^2\bar{V}(x)}{dx^2} + \beta^2\bar{V}(x) = 0$$

with $\beta = \omega\sqrt{LC}$ the *propagation constant*

Solution of the ODE I

The previous equation has a solution of the type

$$\bar{V}(x) = \bar{V}_1 e^{\beta j x} + \bar{V}_2 e^{-\beta j x}.$$

By derivation, the current is

$$\bar{I}(x) = (\bar{V}_1 e^{\beta j x} - \bar{V}_2 e^{-\beta j x})/Z_c.$$

With the **surge impedance**

$$Z_c = \sqrt{\frac{L}{C}}.$$

Solution of the ODE II

The boundary conditions at $x = 0$,

$$\bar{V}(0) = \bar{V}_R = V_R \angle 0,$$

and

$$\bar{I}(0) = \bar{I}_R$$

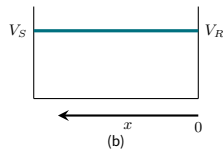
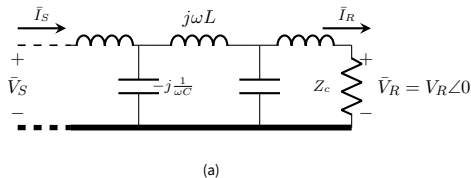
allow to determine constants $\bar{V}_1$ and $\bar{V}_2$, and finally

$$\bar{V}(x) = \bar{V}_R \cos(\beta x) + jZ_c \bar{I}_R \sin(\beta x).$$

Surge impedance loading

Closing the line on the surge impedance Z_c

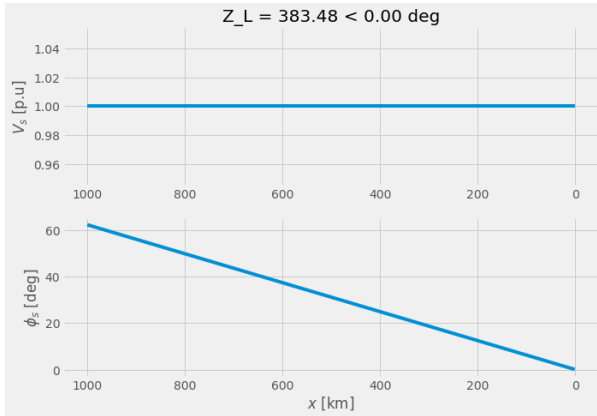
If the line is assumed lossless and we close it with Z_c , assuming $\bar{V}_R = V_R \angle 0$:



then the voltage *magnitude is constant* over the line: $\bar{V}(x) = V_R e^{j\beta x}$, and only the *angle increases with x* . Similar conclusion for $\bar{I}(x)$.

- Why? The reactive power consumed by the line is the same as the reactive power produced, everywhere.

Illustration in Python



SIL, 230 kV line params

See the [Python notebook](#).

Surge impedance loading

Z_c depends on the line characteristics/geometry and is, hence, mainly a function of the voltage level (distances between conductors, etc.).

The surge impedance loading (SIL) is the power drawn by the load Z_c , which depends on the voltage level V_{LL}

$$SIL = \frac{V_{LL}^2}{Z_c}$$

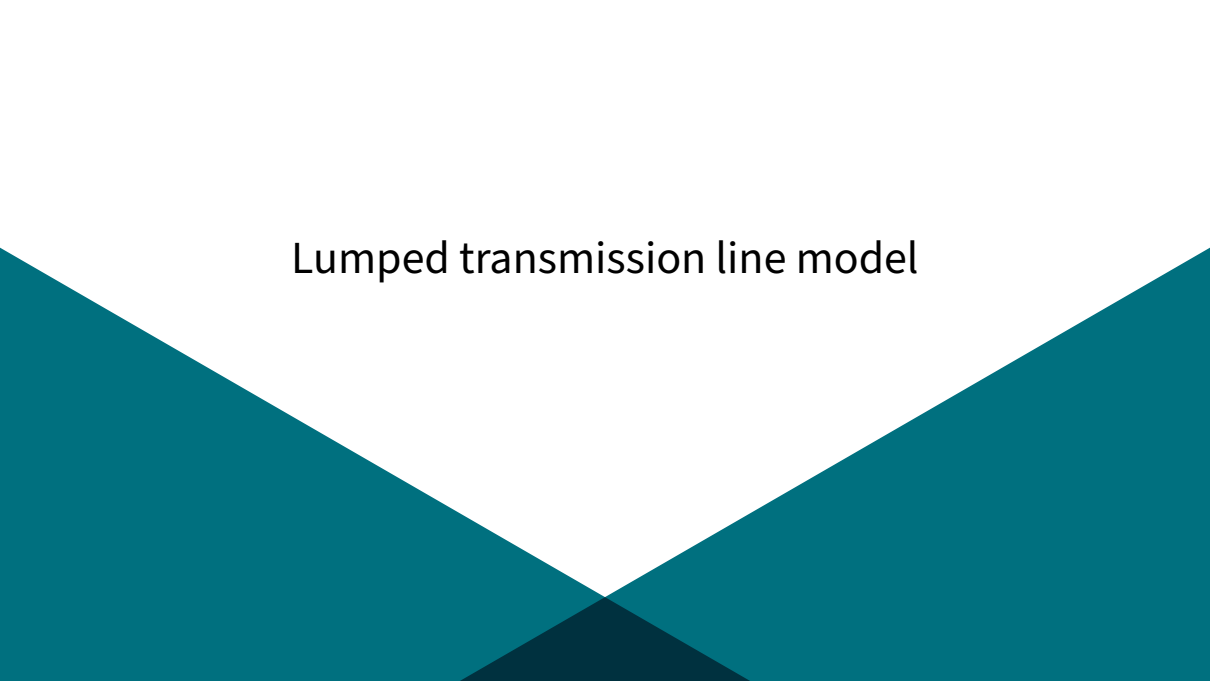
Example: for 500 kV, $SIL \approx 1020$ MW. (Exercise: show it)

Line loadability

The SIL gives an idea of the loadability of a line depending on its length:

- ▶ short line, $l < 100$ km
 - ▶ load limit = $3 \times SIL$
 - ▶ thermal limit (See Section 5 on Line rating)
- ▶ Medium length line, $100 \text{ km} < l < 300 \text{ km}$
 - ▶ load limit = $1.5 \text{ to } 3 \times SIL$
 - ▶ voltage drop $< 5\%$
- ▶ Long line, $l > 300 \text{ km}$
 - ▶ load limit $\approx 1 \times SIL$
 - ▶ for system stability, the angle difference between line ends should stay $< 40^\circ$, see lecture on Transient stability.

Lumped transmission line model

The background of the slide features a white upper half and a teal lower half. The teal section is composed of two large triangles meeting at a point at the bottom center, with a smaller, darker teal triangle positioned directly beneath that point.

The π model

If l is relatively small (< 300 km), we can *approximate* the line with **lumped** parameters:

By manipulation of the previous equations and assuming βl small,

► $Z_{\text{series}} = Rl + j\omega Ll$

► $\frac{Y_{\text{shunt}}}{2} = j\frac{\omega Cl}{2}$

Remember that R , L and C are per km values.

This π model is symmetrical by design.

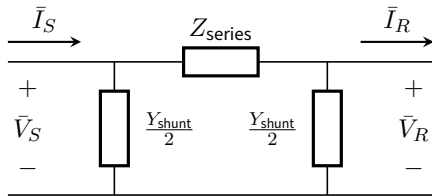
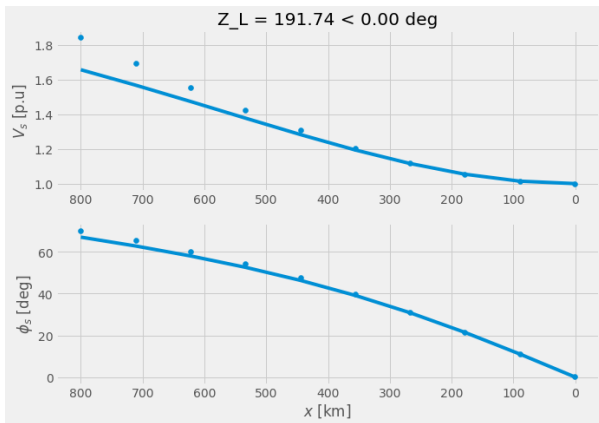


Illustration in Python



230 kV line params, $Z_L = Z_C/2$

Dots are obtained with the π model, while solid lines are from the distributed model. The approximation error grows for $l > 300$ km.

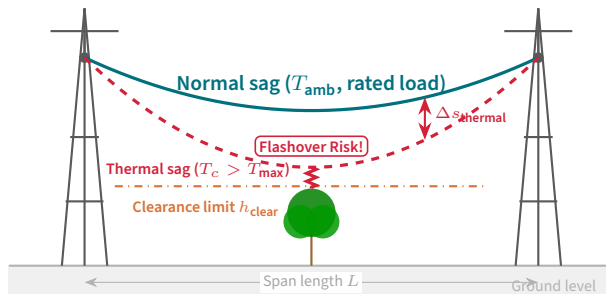
See this [Python notebook](#).

Line rating

Thermal Limits & Ground Clearance I

Transmission lines have a **maximum conductor temperature** $T_{\max}$ (typically 75–90°C):

- ▶ **Thermal elongation:** Current generates Joule losses (RI^2). The conductor expands ($\Delta L = \alpha L \Delta T$), causing sag to increase: $s \approx \frac{wL^2}{8H}$.
- ▶ **Ground clearance violation:** Sag reduces distance to ground/trees $\Rightarrow$ risk of catastrophic **flashover** (major contributor to 2003 US blackout!).
- ▶ **Material annealing:** Sustained excessive temperature permanently degrades aluminum tensile strength.



Thermal Limits & Ground Clearance II

Symbol	Parameter	Unit	Description
s	Sag	m	Vertical drop at mid-span relative to tower support line
w	Linear load	N/m	Weight per unit length ($m_{\text{line}}g$), plus ice/wind loads
L	Span length	m	Horizontal distance between adjacent support towers
H	Tension	N	Constant horizontal component of mechanical tension
α	Thermal coeff.	K^{-1}	Linear expansion coefficient in $\Delta L = \alpha L \Delta T$ ($\approx 15\text{--}23 \times 10^{-6} \text{ K}^{-1}$ for ACSR/Al)

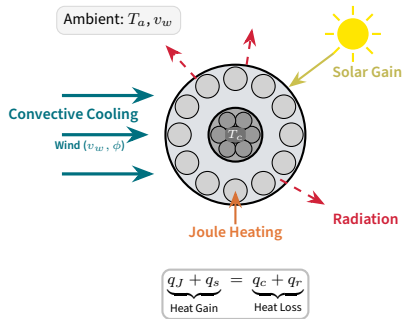
Conductor Heat Balance (IEEE 738 / CIGRE 601)

In steady-state, heat gain balances heat loss: $q_J + q_s = q_c + q_r$.

- ▶ **Joule heating:** $q_J = R(T_c) \cdot I^2$
- ▶ **Solar gain:** $q_s = \alpha_s \cdot S \cdot D$
- ▶ **Convection (wind):** $q_c = h_c(v_w, \phi)(T_c - T_a)$ (**dominant!**)
- ▶ **Radiation:** $q_r = \pi D \epsilon \sigma_B (T_c^4 - T_a^4)$

Thermal ampacity limit at $T_c = T_{\max}$ (net cooling headroom):

$$R(T_{\max}) \cdot I_{\max}^2 = \underbrace{q_c + q_r}_{\text{cooling}} - \underbrace{q_s}_{\text{solar gain}}$$
$$\Rightarrow I_{\max} = \sqrt{\frac{q_c(T_{\max}) + q_r(T_{\max}) - q_s}{R(T_{\max})}}$$



Standards & Heat Balance Parameters

Governing International Standards:

- ▶ **IEEE Std 738** (*IEEE PES*): Standard for calculating current-temperature relationships of bare overhead conductors (predominant in North America and globally).
- ▶ **CIGRE TB 601** (*CIGRE WG B2.43*): Guide for thermal rating calculations of overhead lines (widely used across Europe and by Belgian TSO Elia).

Both standards provide the physical formulations used in EMS software for **SLR** and **DLR**.

Symbol	Electrical / Thermal Flux	Unit	Symbol	Environmental / Line Parameter	Unit
q_J	Joule heat generation (RI^2)	W/m	T_a	Ambient air temperature	°C
q_s	Solar radiation heat gain	W/m	$T_{\max}$	Maximum conductor temperature	°C
q_c	Convective heat loss (wind)	W/m	v_w, ϕ	Wind speed and incident angle	m/s, °
q_r	Radiative heat loss	W/m	D	Conductor outer diameter	m
$I, I_{\max}$	Current / Thermal ampacity	A	S	Global solar irradiance	W/m ²
$R(T_c)$	AC resistance at T_c	Ω/m	α_s, ϵ	Solar absorptivity, emissivity	—

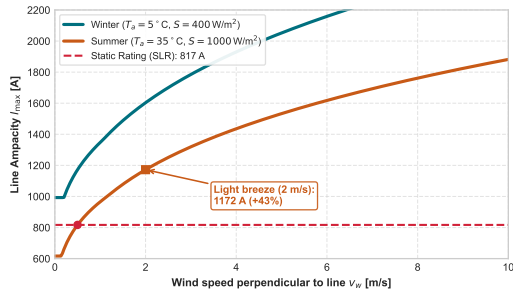
Static (SLR) and Seasonal (SeLR) Line Rating

Static Line Rating (SLR):

- ▶ Conservative worst-case assumption: high T_a (35–40°C), low wind ($v_w \approx 0.5$ m/s), full sun.
- ▶ Constant safety baseline across the year.

Seasonal Line Rating (SeLR):

- ▶ Separate Summer/Winter ratings (+40% base ampacity in winter due to cold ambient air).



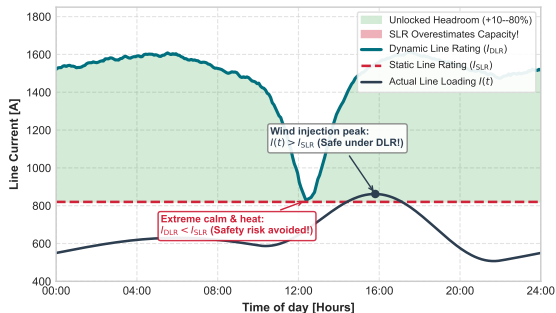
The Opportunity for DLR:

- ▶ Atmospheric cooling is much stronger $> 95\%$ of time.
- ▶ A gentle **2 m/s breeze adds +43% capacity**, but static limits keep this trapped!

Dynamic Line Rating (DLR) & Wind Power Synergy

Dynamic Line Rating (DLR / RTTR):

- ▶ Continuously updates ampacity $I_{DLR}(t)$ from real-time and forecasted weather/line state.
- ▶ **Wind Synergy:** Strong wind generation coincides with high convective cooling: **capacity unlocks when needed most!**



Operational Value:

- ▶ **Headroom:** Unlocks +10~50% capacity on existing lines without new construction.
- ▶ **Safety:** Detects rare calm heatwaves where static ratings overestimate ampacity ($I_{DLR} < I_{SLR}$).

DLR in Practice: Sensors & Grid Integration

Sensing & Monitoring Technologies:

- ▶ **Autonomous line sensors:** Mounted on conductors to track vibration, sag, and temperature (e.g., **Ampacimon ADR**, a ULiège spin-off!).
- ▶ **Tension & sag monitors:** Load cells at dead-end towers.
- ▶ **Weather-based:** Local stations + weather forecasts.

Grid Integration & Industrial Value:

- ▶ Belgian TSO **Elia** pioneered DLR on 150/380 kV lines.
- ▶ Fast live-line drone installation (no power outage).
- ▶ SCADA/EMS integration for dynamic congestion relief.

Video: DLR in Action

[Watch Video \(YouTube\)](#)

Demonstrated in the field:

- ▶ Live-line drone sensor mounting
- ▶ Vibration-based real-time sag tracking
- ▶ Real-time dynamic ampacity dispatch